Smart Traffic Accident Analysis and Severity Prediction

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Introduction

- **Problem:** road accidents block traffic and slow emergency response; estimating early how much an accident will disrupt traffic helps target police and recovery crews.
- **Data:** US Accidents (Moosavi et al., Kaggle): 7,728,394 accident records, 2016-01-14 to 2023-03-31, with GPS, weather, time and road-feature attributes.
- **Objectives:** clean the records and handle missing GPS values; measure weather influence; find accident hotspots; run correlation analysis; predict crash Severity (1–4).
- **Scope:** five modules: Data Science Concepts, Exploratory Data Analysis, Data Visualization, Machine Learning, Big Data Analytics.

Methods

- **Ingest:** 3.06 GB CSV read in chunks; stratified sample of 399,999 rows (random_state = 42).
- **Clean & features:** duplicates and impossible values removed, outliers winsorized, End GPS imputed from Start, weather by State + month.
- **Models:** Decision Tree, Random Forest, Logistic Regression, HistGradientBoosting; class-balanced, stratified 80/20 split, CV tuning.
- **Big data:** MongoDB (2dsphere) and Hadoop-style MapReduce, cross-checked.



CSV → Parquet → MongoDB / MapReduce → ML → Streamlit dashboard

Results

Key findings

- **Hotspots (accidents per 0.1° cell):** Los Angeles, CA (50,254), Miami, FL (42,955), Charlotte, NC (36,259), Baton Rouge, LA (33,612)
- **Weather:** Rain has the highest severe share (22.9%), Fog/Haze the lowest (15.8%); weak link (Cramér's V = 0.039).
- **Time pattern:** Peak hours are 07:00 and 16:00; a weekend day averages 53% fewer accidents than a weekday.
- **Predictors:** Start_Lng, City and Start_Lat lead the permutation importance; Year excluded as a collection artifact.
- **Missing GPS:** no Start GPS was missing; 176,088 End coordinates (44.1% of 399,581 records) were imputed from the Start point.

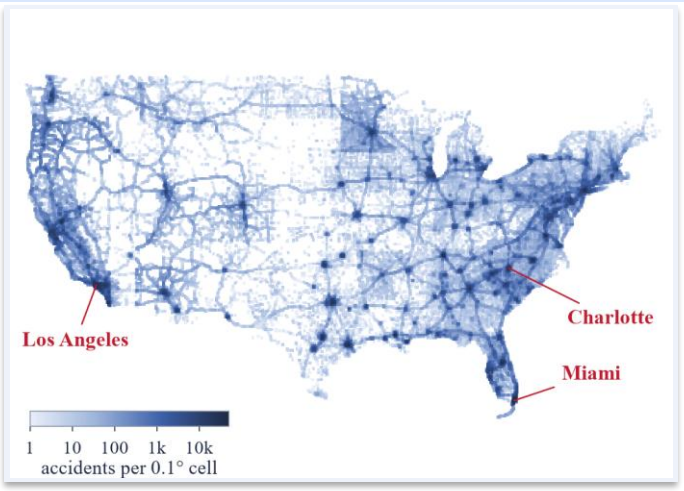


Fig. 1 Accident density, all 7.7 million records

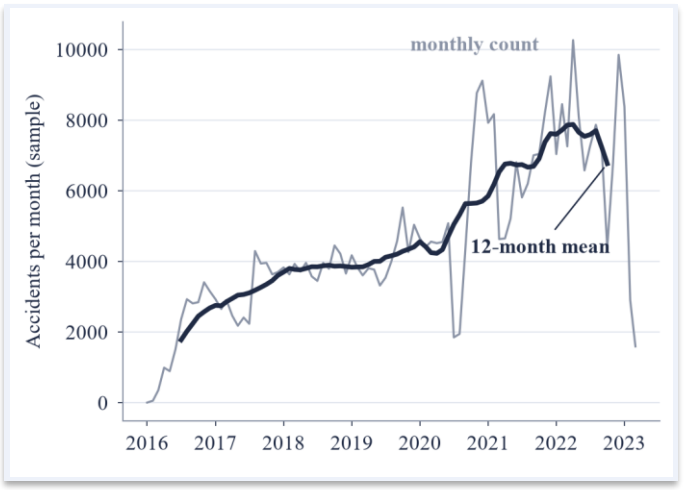


Fig. 2 Monthly accidents, 12-month mean

Model comparison (test set)

Model	Accuracy	Macro-F1	ROC-AUC
Decision Tree	64.9%	0.353	0.627
Random Forest	59.5%	0.375	0.810
Logistic Regression	37.4%	0.261	0.735
HistGradientBoosting	**59.0%**	**0.381**	**0.828**
Majority baseline	79.7%	0.222	n/a

- **MongoDB:** 399,581 documents, 2dsphere index; 10 queries timed with / without indexes.
- **MapReduce:** 4 splits, 4 maps, 2 reducers; all 3 jobs match pandas and MongoDB.

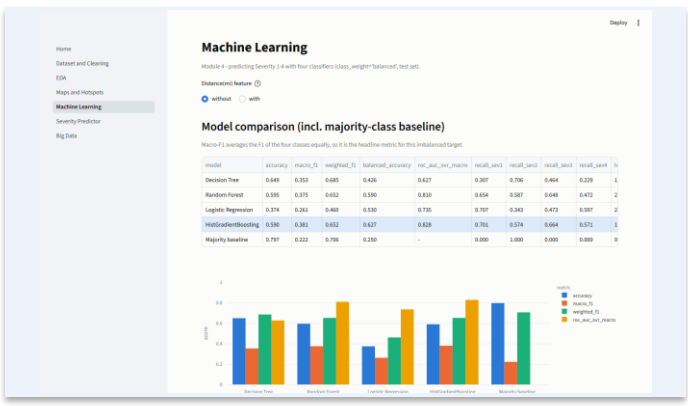


Fig. 3 Streamlit dashboard (ML page)

Discussions

Conclusion

- HistGradientBoosting gave the best result (macro-F1 0.381, accuracy 59.0%).
- All models fall below the 79.7% baseline accuracy, so macro-F1 is used; it finds 66% of Severity 3 and 57% of Severity 4 accidents.

Limitations

- Severity classes are imbalanced, so macro-F1 is reported next to accuracy.
- Models were trained on a sample, not on all records.
- Weather is taken from the nearest weather station, so it is approximate.

Future work

- Real-time severity alerts from streaming data.
- Spark MLlib on all records instead of a sample.
- Add road-network and traffic-volume features.

Scan for the code and results

Dataset: Moosavi et al., US Accidents (Kaggle), CC BY-NC-SA 4.0. Code: github.com/Abin-Shaji-Thomas/Smart-Traffic-Accident-Analysis-and-Severity-Prediction